INCH-POUND

MIL-DTL-55302/147C
w/AMENDMENT 1
16 March 2011
SUPERSEDING
MIL-DTL-55302/147C
22 March 2005

DETAIL SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, RECEPTACLE WITH PIN CONTACTS, 2 FEMALE KEYING GUIDES, 3.315 INCHES LENGTH, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

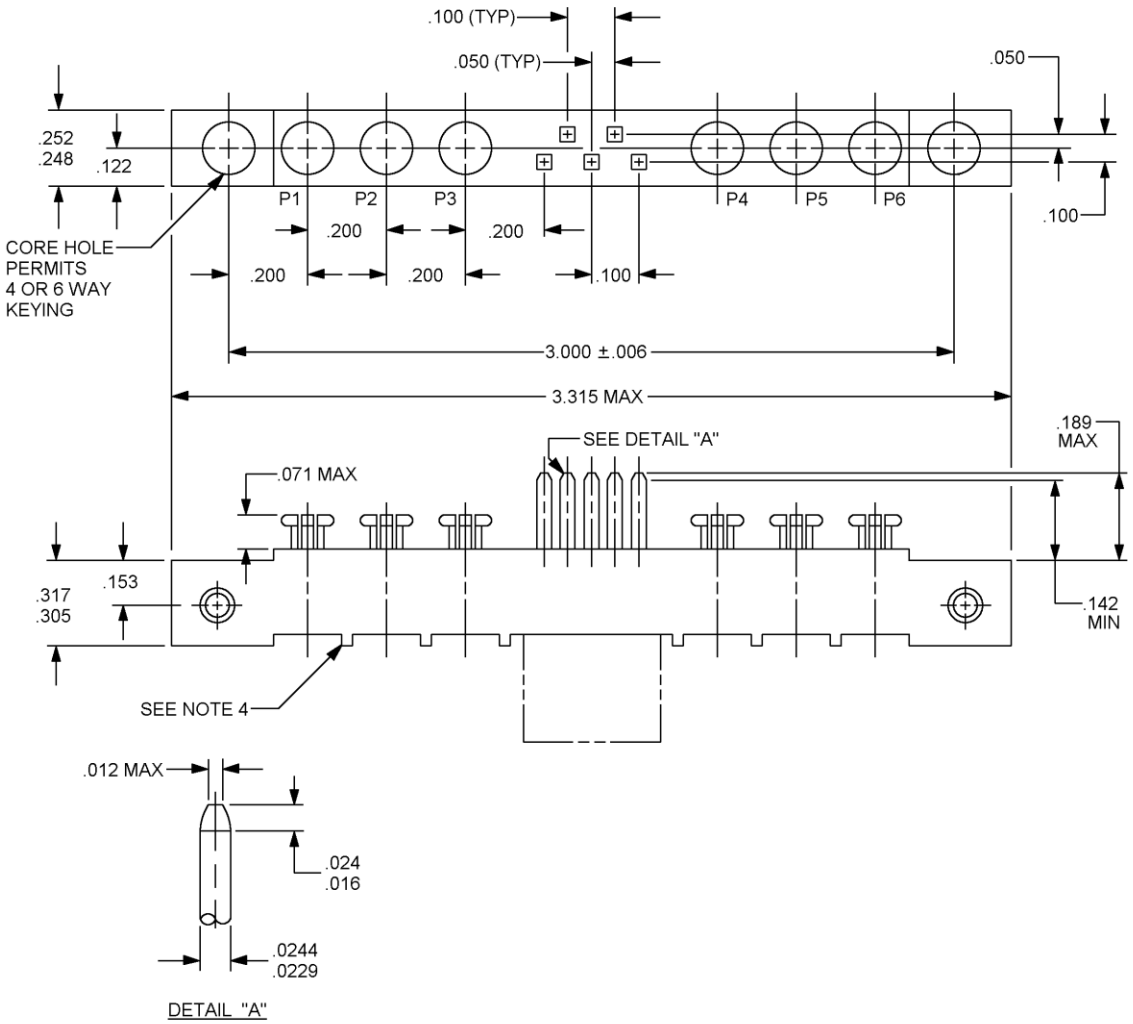



FIGURE 1. Connector, plug (.100 inch spacing).

AMSC N/A FSC 5935

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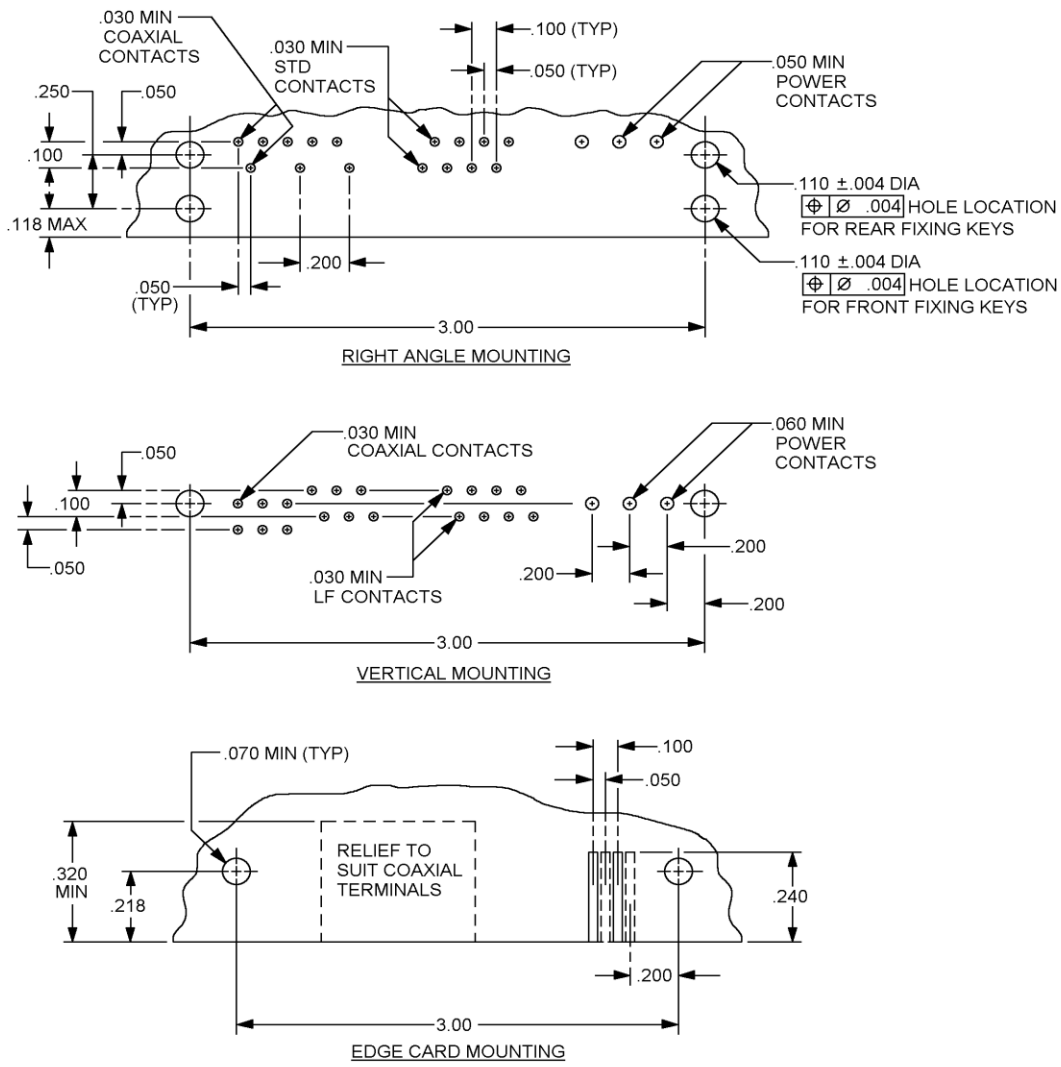
| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .006 | 0.15 | 0.071 | 1.80 | .248 | 6.30 |
| .012 | 0.30 | 0.100 | 2.54 | .252 | 6.40 |
| .016 | 0.41 | 0.122 | 3.10 | .305 | 7.75 |
| .0229 | 0.582 | 0.142 | 3.61 | .317 | 8.05 |
| .024 | 0.61 | 0.153 | 3.89 | 3.000 | 76.20 |
| .0244 | 0.620 | 0.189 | 4.80 | 3.317 | 84.25 |
| .050 | 1.27 | 0.200 | 5.08 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. Standoffs are located between each cable hole and between each hole and contact hole set.

FIGURE 1. Connector, plug (.100 inch spacing) - Continued.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .004 | 0.10 | .100 | 2.54 | .218 | 5.54 |
| .030 | 0.76 | .106 | 2.69 | .240 | 6.10 |
| .050 | 1.27 | .110 | 2.79 | .320 | 8.13 |
| .060 | 1.52 | .118 | 3.00 | .640 | 16.26 |
| .070 | 1.78 | .200 | 5.08 | 3.00 | 76.2 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).

FIGURE 2. Recommended hole patterns.



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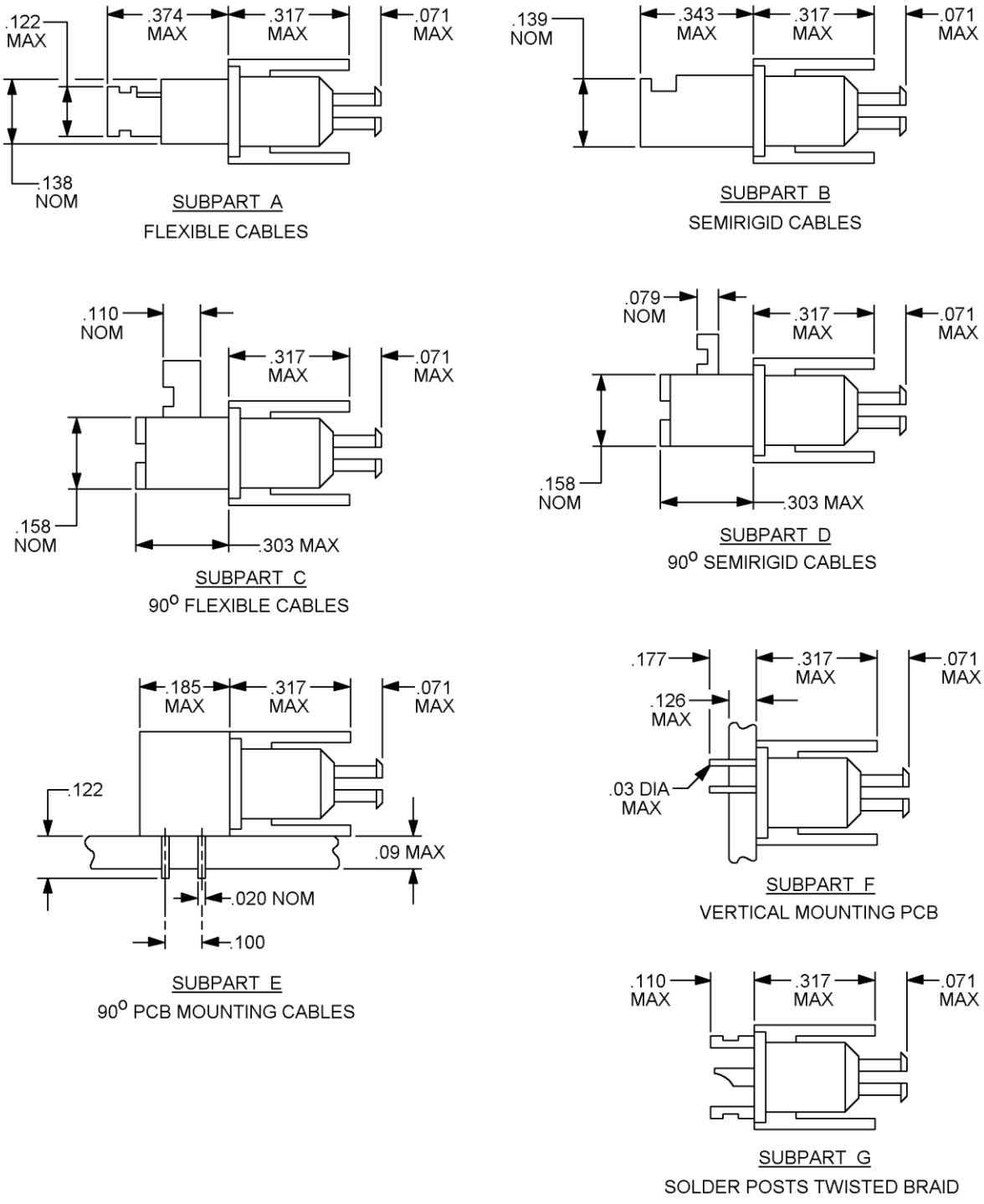
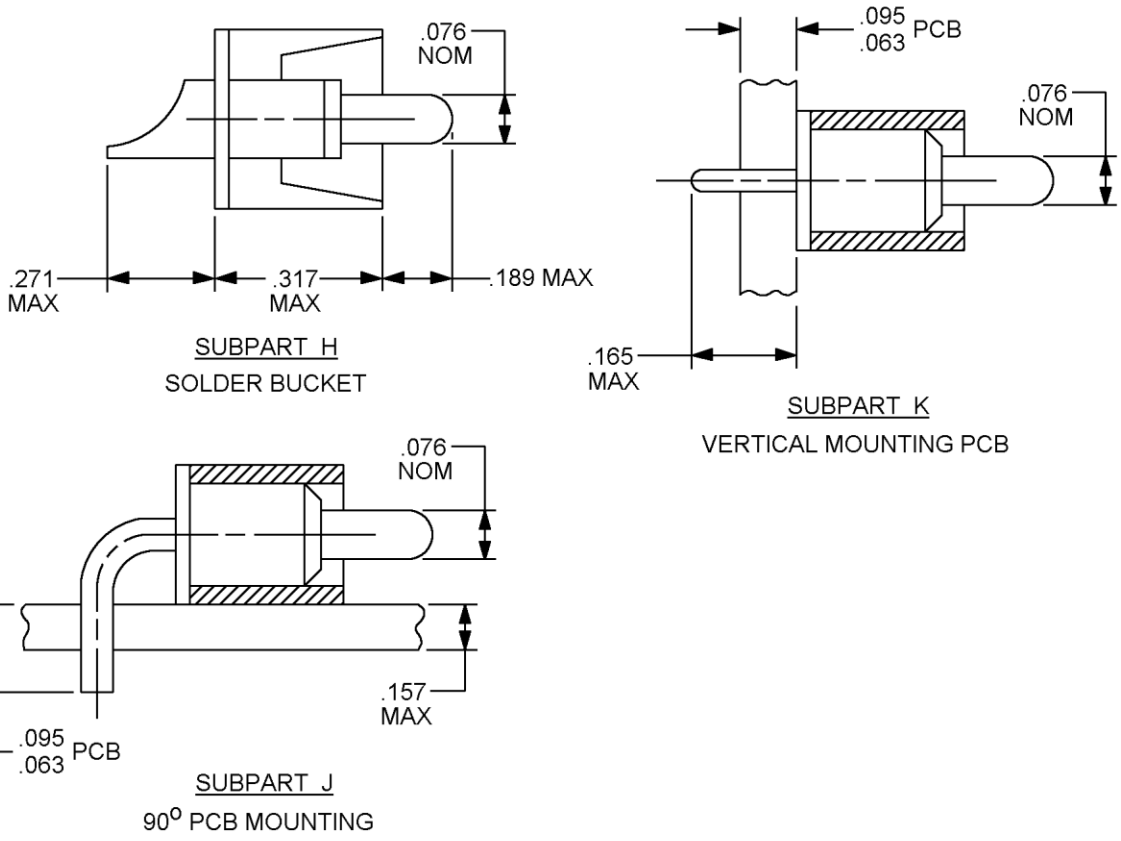



FIGURE 3. Coaxial configurations.



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| Inches | mm | Inches | mm |
|---|---|---|---|
| .020 | 0.51 | .139 | 3.53 |
| .03 | 0.8 | .155 | 3.94 |
| .063 | 1.60 | .157 | 3.99 |
| .071 | 1.80 | .158 | 4.01 |
| .076 | 1.93 | .165 | 4.19 |
| .079 | 2.01 | .177 | 4.50 |
| .094 | 2.39 | .185 | 4.70 |
| .095 | 2.41 | .189 | 4.80 |
| .100 | 2.54 | .271 | 6.88 |
| .110 | 2.79 | .303 | 7.70 |
| .122 | 3.10 | .317 | 8.05 |
| .126 | 3.20 | .343 | 8.71 |
| .138 | 3.51 | .374 | 9.50 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).

FIGURE 3. Coaxial configurations – Continued.



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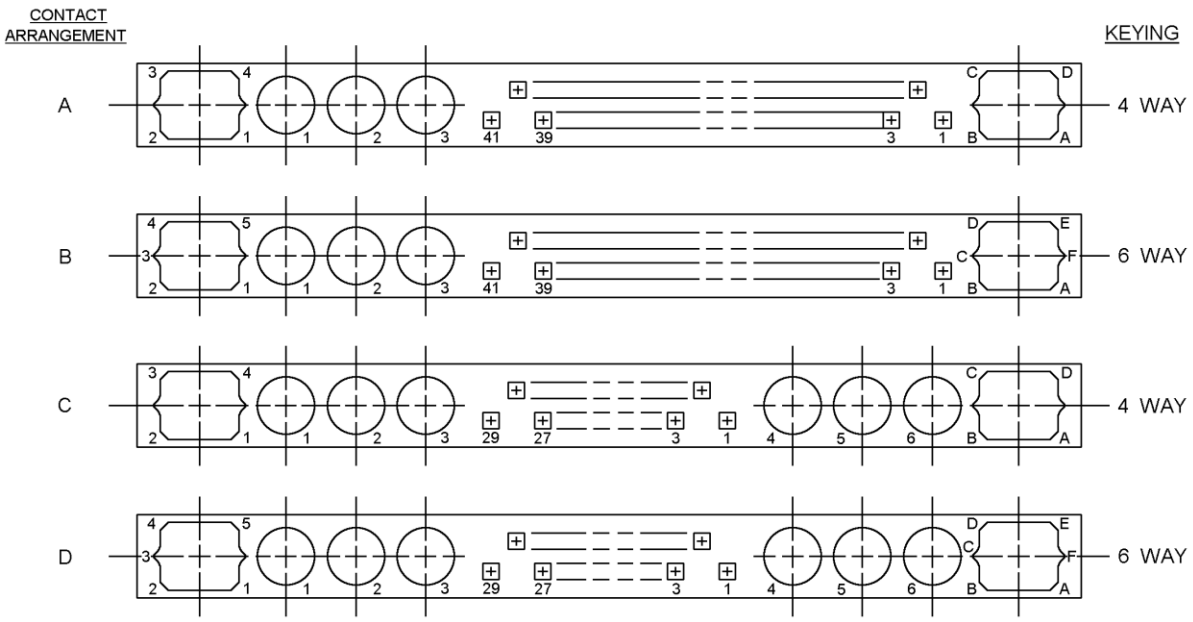



FIGURE 4. Contact and keying arrangements.



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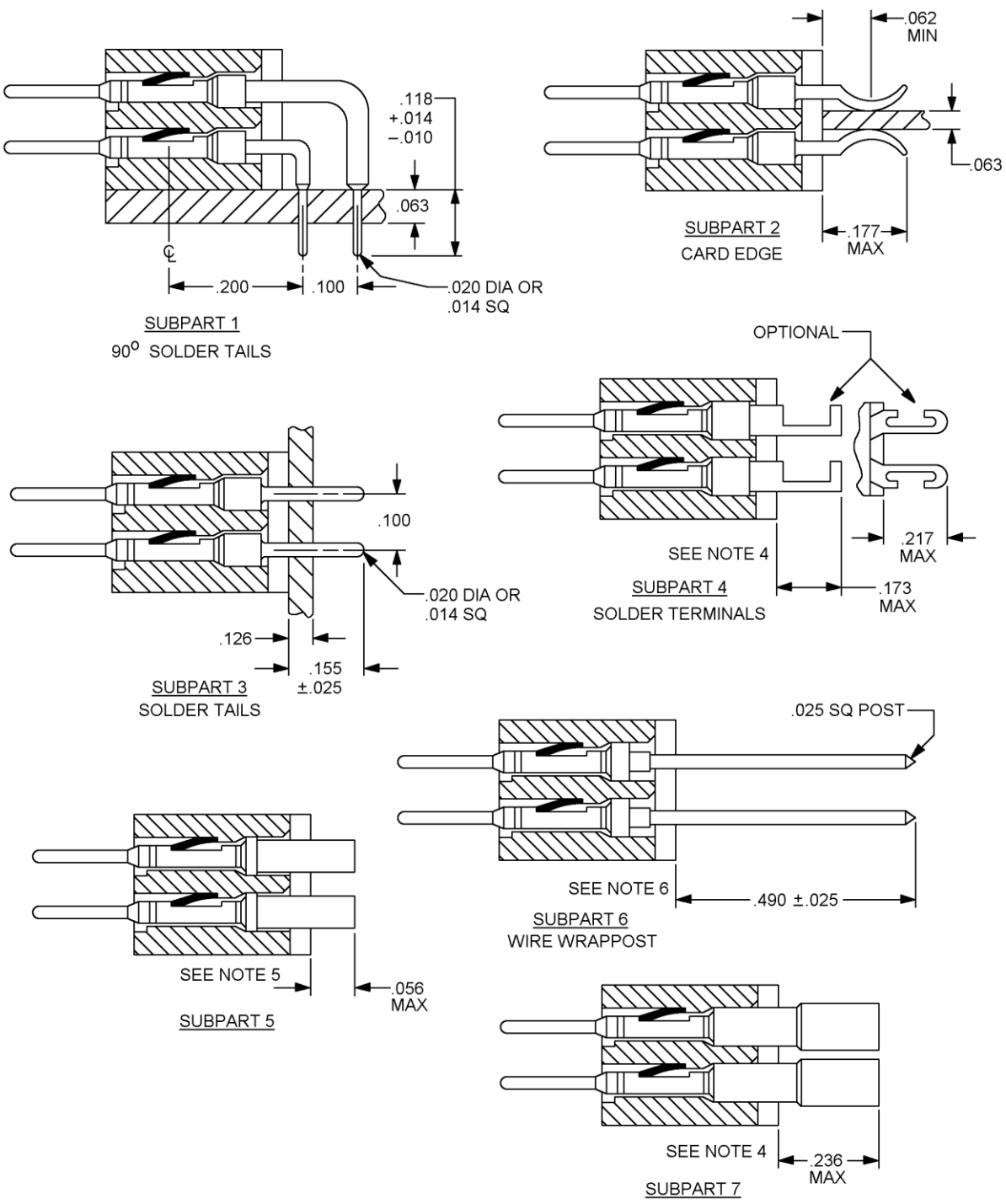



FIGURE 5. Connector terminations.



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| Inches | mm | Inches | mm |
|---|---|---|---|
| .010 | 0.25 | .126 | 3.20 |
| .014 | 0.36 | .155 | 3.94 |
| .020 | 0.51 | .173 | 4.39 |
| .025 | 0.64 | .177 | 4.50 |
| .056 | 1.42 | .200 | 5.08 |
| .062 | 1.57 | .217 | 5.51 |
| .063 | 1.60 | .236 | 5.99 |
| .100 | 2.54 | .490 | 12.45 |
| .118 | 3.00 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is $\pm$ .005 (0.13 mm).
4. Solder tail accepts no. 22 AWG wire.
5. Crimp contact accepts no. 22-26 AWG wire.
6. Contact with .025 inch$^2$ (0.64 mm$^2$) wrappost.

FIGURE 5. Connector terminations - Continued.



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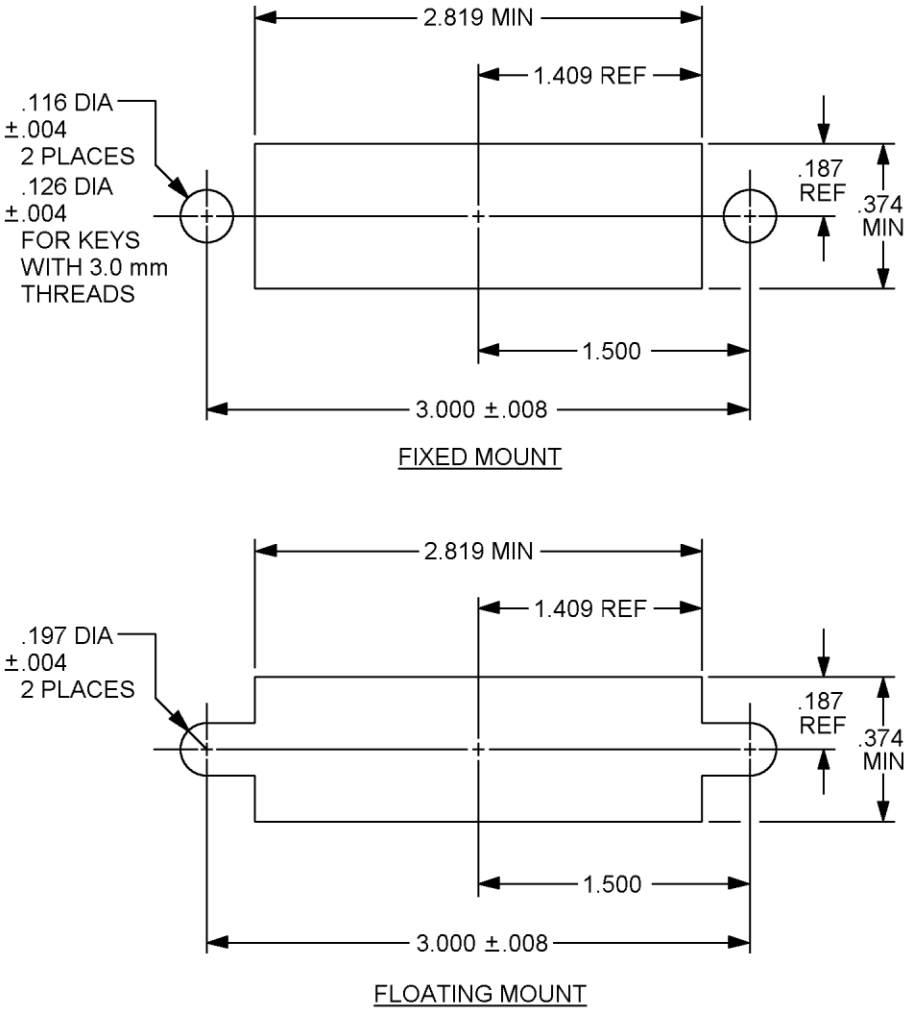



FIGURE 6. Recommended panel cutouts.



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| Inches | mm | Inches | mm |
|---|---|---|---|
| .004 | 0.10 | .374 | 9.50 |
| .008 | 0.20 | 1.409 | 35.79 |
| .116 | 2.95 | 1.500 | 38.10 |
| .126 | 3.20 | 2.819 | 71.60 |
| .187 | 4.75 | 3.000 | 76.20 |
| .197 | 5.00 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. Floating mount panel thickness .085 (2.16 mm) maximum.

FIGURE 6. Recommended panel cutouts - Continued.



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REQUIREMENTS

Dimensions and configurations: See figures 1 through 6.

Contact identification: See figure 4.

Contact retention: Coaxial and power: 11.25 pounds minimum.
Signal: 4 pounds minimum.

| Mating and unmating force (per contact): | Coaxial | Power | Signal |
|---|---|---|---|
| (Min) | 7.2 ounces | 3.6 ounces | 0.7 ounce |
| (Max) | 36 ounces | 18 ounces | 3.6 ounces |

Values for each connector to be addition of contacts according to contact type.

Contact resistance: 12 milliohms maximum.

Dielectric withstanding voltage:

Sea level: 1,000 V rms, 60 Hz.

High altitude: 300 V rms, 60 Hz.

Current rating:

Power contacts: 25 amperes at 25°C.

Coaxial contacts: .5 ampere at 25°C.

Signal contacts: 3 amperes at 25°C.

Keying: Moldings and mounting hardware provide 16 or 36 polarizing positions according to choice of 4 or 6 way capability. Keys are constructed in accordance with MIL-DTL-55302/154. Connectors are supplied with hex pins in the “1” and “A” positions, except for 1/4-turn locking styles, which are keyed in “F” and “6” positions. When required, connectors can be easily keyed to desired position by using a simple wrench.

Hood types: In accordance with MIL-DTL-55302/155.

Mounting hardware: In accordance with MIL-DTL-55302/154.

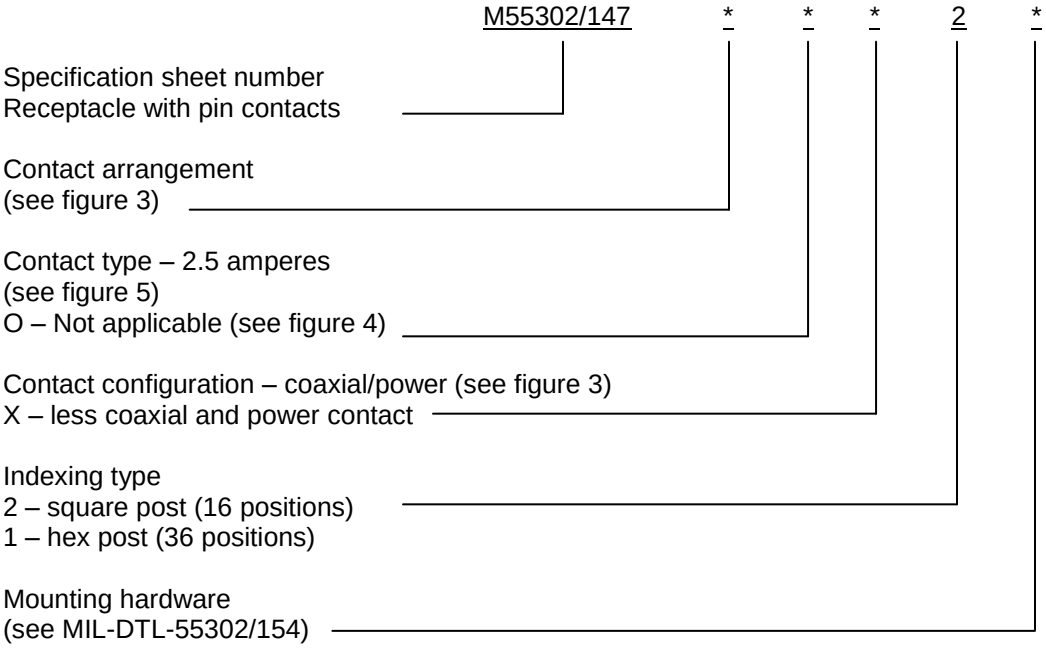
Mating connectors: In accordance with MIL-DTL-55302/152.



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Part or Identifying Number (PIN): Consists of the letter M, the basic number of the specification sheet, and a dash number compiled as follows:

Example:




Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/152
MIL-DTL-55302/154
MIL-DTL-55302/155



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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2010-124)

Review activities:
Army – AR, AT, AV, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

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